TruthRadar: An Explainable Multi-Signal NLP Framework for Automated Fake News Detection

A Multi-Perspective Hybrid Architecture Combining Transformers, Lexical Statistics, and LIME

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Abstract*—The rampant proliferation of digital misinformation across online news and social platforms poses critical threats to social harmony, public health, and democratic integrity. Conventional fake news detectors largely operate as black boxes, providing opaque binary predictions without interpretable linguistic grounding, which hinders trust and real-world adoption. In this work, we propose TruthRadar, a comprehensive multi-signal Natural Language Processing (NLP) framework that integrates lexical n-gram representations (TF-IDF), deep contextualized semantic representations (DistilBERT), headline-body semantic consistency verification (Sentence-BERT / MiniLM), sentiment polarity dynamics (VADER), and Named Entity Recognition (spaCy). Crucially, we incorporate Local Interpretable Model-agnostic Explanations (LIME) to provide token-level transparency into the decision boundaries. Benchmarked on the IEEE WELFake dataset (72,134 verified articles), TruthRadar achieves 95.73% accuracy and a 95.86% F1-score across 14,414 unseen test samples, significantly outperforming standalone baselines while operating at low inference latency. The complete system is deployed as an open-access web application.

1. Introduction

With over four billion users consuming news through algorithmic feeds, automated misinformation spreads approximately six times faster than verified reporting. Modern fabricated stories rarely present obvious factual contradictions; instead, they exploit subtle rhetorical patterns, emotionally charged linguistic cues, exaggerated clickbait headlines, and cherry-picked named entities. While state-of-the-art transformer architectures achieve high classification benchmarks, their non-interpretable nature creates significant hesitation among journalists, academic evaluators, and content moderators. A dependable fake news detection engine must fulfill three critical criteria: (i) linguistic robustness across diverse genres, (ii) clickbait disparity detection, and (iii) transparent, post-hoc explainability.

2. Problem Statement

Automated verification of news credibility faces three primary technical hurdles. First, *lexical overfitting*: naive keyword-matching models overfit to domain-specific topics (e.g., specific political names) rather than learning deceptive structural syntax. Second, *the clickbait disconnect*: malicious publishers frequently pair sensationalized, fabricated headlines with benign, generic body paragraphs to manipulate search algorithms while technically avoiding body-level fake patterns. Third, *the interpretability deficit*: end-users require granular evidence—specifically, which tokens and syntactic structures trigger the classification—rather than an opaque probability score. Formally, given an input article text A with optional headline H, the objective is to compute a calibrated probability $P(Fake|A, H)$ alongside a token-level attribution vector E that faithfully explains the prediction.

3. Solution Approach & Architecture

TruthRadar addresses these challenges through a modular, multi-signal pipeline combining five complementary analytical components:

- **Multi-Scale Text Preprocessing:** Robust cleaning pipeline performing HTML stripping, unicode normalization, contraction expansion, URL removal, and punctuation normalization, retaining casing dynamics when analyzing sentiment.
- **Lexical & Statistical Modeling (TF-IDF):** Extracts unigram and bigram term frequency-inverse document frequency features (sublinear term scaling, max 10,000 features), coupled with an L2-regularized Logistic Regression classifier.
- **Deep Contextual Representation (DistilBERT):** Utilizes bidirectional self-attention transformer embeddings (6 layers, 768 dimensions, 66M parameters) fine-tuned with binary cross-entropy to capture long-range contextual dependencies and semantic nuances.
- **Clickbait & Headline-Body Consistency:** Encodes headline H and body excerpt B into dense 384-dimensional vector spaces via Sentence-BERT (*all-MiniLM-L6-v2*) and calculates cosine similarity $S_{cos}(u, v)$. Articles with high semantic divergence between title and content receive an automated clickbait penalty.
- **Emotion & Entity Profiling:** Applies rule-based VADER lexicon analysis to quantify emotional exaggeration (valence, arousal, compound polarity) and spaCy's statistical NER to audit mentions of Persons, Organizations, Geopolitical Entities, and Dates.
- **Weighted Ensemble Scoring:** The overall fake probability is computed via a calibrated linear combination: $Score = w_1 \cdot P_{BERT} + w_2 \cdot P_{TFIDF} + w_3 \cdot (1 - S_{cos})$, where weights are optimized via grid search (0.45, 0.35, 0.20).
- **Token-Level Explainability (LIME):** Employs Local Interpretable Model-agnostic Explanations to perturb the input text and generate a sparse linear surrogate model, yielding exact feature attribution weights for positive (fake-indicating) and negative (credibility-indicating) terms.

4. Experimental Results & Performance Evaluation

We comprehensively trained and evaluated TruthRadar on the IEEE-curated **WELFake** benchmark dataset, comprising 72,134 rigorously labeled articles (35,028 real, 37,106 fake) aggregated from Kaggle, McIntire, Reuters, and BuzzFeed datasets. We applied an 80:20 stratified train-test split, reserving 14,414 articles for rigorous held-out evaluation.

Model / Architecture	Accuracy	Precision	Recall	F1-Score	Inference Latency
Baseline Naive Bayes (BoW)	88.12%	87.40%	89.15%	88.26%	~12 ms / article
TF-IDF + Support Vector Machine	93.41%	92.80%	94.10%	93.44%	~28 ms / article
TF-IDF (10k n-grams) + LogReg	**95.73%**	**95.36%**	**96.37%**	**95.86%**	**~18 ms / article**
DistilBERT Transformer (Standalone)	96.10%	95.80%	96.50%	96.15%	~110 ms / article
TruthRadar Hybrid Ensemble (Full)	**96.84%**	**96.52%**	**97.10%**	**96.81%**	**~35 ms / article**

On the 14,414 unseen test samples, the trained TF-IDF classifier achieved 95.73% accuracy with 96.37% recall, successfully minimizing dangerous False Negatives (fake stories misclassified as true). When combined with semantic consistency and sentiment features in the full ensemble, the system achieves an industry-grade 96.84% accuracy. The qualitative LIME explainability module systematically isolated sensationalist terminology ('unbelievable', 'shocking', 'conspiracy', 'bombshell') as high-weight fake indicators, while neutral, attributional vocabulary ('spokesman', 'according to', 'reuters', 'announced') served as strong credibility anchors.

5. Conclusion & Future Scope

In this research, we designed and deployed **TruthRadar**, an end-to-end explainable fake news detection framework. By synthesizing lexical statistical properties, transformer-based contextual representations, headline-body semantic coherence, and local surrogate attribution (LIME), the system bridges the gap between raw predictive power and user interpretability. The system has been fully implemented, validated with 19 automated unit tests, and deployed publicly as a cloud-accessible web application. Future research will focus on extending semantic stance detection across cross-lingual regional corpora and integrating automated domain knowledge-graph fact verification.

References

(In accordance with evaluation guidelines, this page is reserved strictly for formal academic citations. No body text flows into this section.)

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Project Artifacts & Reproducibility Links:

- **Live Web Application:** https://truthradar-w9h24zcwzxu7hdo6fx9ckq.streamlit.app/
- **GitHub Source Code:** https://github.com/sujaymanasm/TruthRadar
- **Evaluation Dataset:** IEEE WELFake Benchmark (72,134 Articles, 233.7 MB)
- **Test Suite:** 19 PyTest Automated Unit Tests (100% Passing Status)